

EDUCATION

- **Indian Institute Of Technology Kanpur** Kanpur, India
B.Tech - Mechanical Engineering; Minor - Machine Learning & Applications 2021 - 2025
Courses: Data Structure & Algorithms, Probabilistic Machine Learning, Optimization Theory, Stochastic Processes & Applications, Probability & statistics, Image Processing, Linear Algebra & ODE, Game Theory & Applications, Robotics, Machine Design

WORK EXPERIENCE

- **Shiprocket Ltd.** Gurugram, India
Graduate Engineer Trainee - Software Engineer Aug 2025 - Present
 - **Built & Designed** an end-to-end **multi-agent** analytics platform using **FastAPI** and **LangGraph**, enabling teams and sellers to execute natural language queries, text synthesis, data visualizations, and predictive forecasting
 - **Designed** a 2-layer **retrieval engine** integrating a pre-computed Platinum KPI database layer with a Trino query engine fallback, utilizing **pgvector** for intent-based question caching, with process logs to **minimize LLM system latency**
 - Engineered an autonomous application **health monitoring system** that processes CubeAPM error logs via webhooks, feeding relevant **codebase context** into an iterative LLM troubleshooting loop to isolate root causes and propose fixes
 - **Optimized** codebase indexing **cost** by implementing **Tree-sitter** for syntactic, function-level code chunking and constructing a **Merkle tree hash** structure to enable smart, incremental file updates and deletions inside a **Milvus DB**
 - **Optimized** Metabase **latency** by implementing viewport-based **lazy loading** to eliminate redundant dashboard card queries, significantly **reducing load** on Trino, while integrating an optimized, cached LLM dashboard summarization tool
 - **Developed** high-throughput backend pipelines in **Golang**, including a **webhook** for streaming analytics data directly into **Kafka** and a formatting engine for syncing transaction data to the enterprise ERP
- **Maruti Suzuki India Limited** Jun 2024 - Aug 2024
Data Science Intern
 - **Designed** an **NLP** and **RAG** pipeline capable of generating responses based on both structured and unstructured data
 - Utilized NLP techniques like **tokenization**, **lemmatization**, and stop words removal for clean and normalized data
 - Utilized **Langchain** text-splitter, and **HuggingFace** Embeddings and **FAISS** vector database and implemented RAG
 - Researched for the potential **risks** to LLM models and developed **strategies** to mitigate those risks and to evaluate them

RESEARCH EXPERIENCE

- **RHO-LOSS based BNN** 
Prof. Piyush Rai, IIT Kanpur Jan 2025 - Apr 2025
 - Implemented RHO-LOSS strategy, a prioritised data selection strategy, to accelerate DNN training by **18x** while improving test accuracy by **3.56%** on CIFAR-10 and **1.78%** on CIFAR-100
 - Transformed a pre-trained **ResNet-18** into a **Bayesian Neural Network** by replacing normalization layers with Bayesian variants, reducing predictive uncertainty (NLL improved from **4.07** to **1.93**)
 - Developed an end-to-end pipeline combining RHO-LOSS, ABNN and ODD Detection for robust deep learning
 - Achieved **89.23%** accuracy on CIFAR-10 and **57.78%** on CIFAR-100 using RHO-LOSS, outperforming baseline methods
- **Snake Robot Motion Planning** 
Prof. Keval S Ramani, IIT Kanpur Jan 2024 - Apr 2024
 - Developed a **mathematical model** for locomotion of a snake robot of **N-degrees** of freedom made of **B-spline** curves
 - Implemented principles of **Lagrangian Mechanics** and **DH Transformation** to obtain the equations of motion
 - Implemented **NURBs** library and **optimization** techniques to minimize torque required for efficient locomotion
 - Performed state-space transformation to streamline the problem-solving, enhancing computational efficiency

PROJECTS

- **Hangman API Problem** 
Trexquant Investment LP, India Mar 2024 - Apr 2024
 - Utilized **N-gram** frequency analysis for calculating letter probabilities from a corpus of **250,000** training words
 - Employed **dynamic substring matching** and weighted scoring system to balance probabilities across N-gram lengths
 - Designed and implemented algorithm achieving a success rate of 62.5%, outperforming the benchmark target by **12.5%**

SKILLS

- **Languages:** Python, C, C++, Golang, SQL, L^AT_EX
- **Frameworks & Libraries:** Scikit-learn, TensorFlow, Keras, Pandas, PyTorch, OpenCV, FastAPI, LangGraph
- **Tools & Platforms:** GIT, GitHub, n8n, MilvusDB, Apache Flink, Docker, Linux, Cursor, Postman, Claude

HONORS AND AWARDS

- Ranked among **Top 2%** in **International Quant Championship 2023**, conducted by WorldQuant Brain
- Selected as one of **72 finalist teams** out of **15,000+** applicants for **PIWOT 2025**

VOLUNTEER EXPERIENCE

- **Company Coordinator, Placement Office IIT Kanpur** Jun 2022 - Apr 2023
 - Coordinated in a **3-tier team** to conduct the Internship drive of **1000+** and Placement drive of **1400+** students
 - Developed liaisons with **80+** Corporate Teams to invite them to participate in the Internship and Placement drive
- **Academic Mentor, ESC101 IIT Kanpur** Sep 2022 - Jul 2023
 - Conducted online and offline Institute level and hall level remedial classes for course **ESC101** attended by **500+** students
 - Taught programming concepts like **Array**, **Pointers**, **Recursion**, **Linked Lists**, **Stack**, **Queue** in C language